Module 06 Lab Exercise - Contribution Journal

Keira Burdine

I worked with the Titanic dataset, prepared the data, defined the features and target variables, and split the dataset into training and testing portions. I then created, trained, and evaluated a linear regression model to predict passenger ticket fares.

For the classification portion, I converted the sex column into numeric values, selected age, passenger class, and sex as predictors, and used survival as the target. I created and trained a logistic regression model, generated predictions, and evaluated its performance using accuracy_score. The model returned a survival prediction accuracy of 74.83%.

I also completed the Knowledge Check questions. I explained the difference between regression and classification, ran code to display the linear regression coefficients, interpreted what those coefficients meant, and explained why mean squared error and accuracy are appropriate for different prediction tasks.

Throughout the lab, I checked my code, corrected spelling errors that caused execution failures,
reran the cells, and verified the results. I also adjusted the Markdown formatting so my written responses displayed correctly. My contribution included coding, troubleshooting, interpreting the model results, answering the questions, and saving the completed lab as a PDF for submission. This work strengthened my understanding of the complete process from data preparation through model evaluation.

John Grubbs

I completed and ran my own copy of the Module 6 Regression and Classification Models lab using the Titanic dataset. For the regression part, I removed rows with missing ages and used Age and Pclass to predict passenger fares. I split the data into training and test sets, trained a Linear Regression model, and tested its predictions.

The model had a mean squared error of about 3364.92. The square root of that was about $58.01, which showed the average size of the prediction error. I also looked at the coefficients to see how age and passenger class affected the predicted fare.

For the classification part, I changed the Sex column into numbers and used Age, Pclass, and Sex to predict whether a passenger survived. The Logistic Regression model had about 74.83% accuracy. I also answered the knowledge check in my own words.

Luis Adrian

My contribution in this lab was completing the code needed to train and evaluate the regression and classification models. I worked through the data preparation steps, including removing rows with missing ages and converting the Sex column into numerical values. I also used the training and testing sets to keep the data used for learning separate from the data used for evaluation.

I created the LinearRegression and LogisticRegression models, trained them with the appropriate features, and generated predictions on the test data. I evaluated the predictions using **mean_squared_error** and calculated its square root to understand the error in the same units as Fare. For survival prediction, I used **accuracy_score** and obtained **74.83%**. I also completed the knowledge check questions to explain the differences between the models, their coefficients, and their evaluation metrics.